CSP-S 2025 (Senior)

2025-10-25, 14:30–18:30 (4 hours)

2025 CCF Certified Software Professional, CSP-J/S 2025, second round, Senior level (CSP-S). Four problems, in contest order. Each problem reads from and writes to the named files. Time and memory limits are those shown on the QOJ problem pages.

Problem 1: Club Recruitment (社团招新, `club`) QOJ 14597

Time limit: 1 s. Memory limit: 512 MB. Total points: 100.

Problem description

Little L is a member of the school's algorithm society. In this year's school club recruitment, Little L recruited n new members in total, where n is even. Now Little L wants to assign them to different departments of the society.

The algorithm society has three departments. The satisfaction of the i-th ($1 \le i \le n$) new member with the j-th ($1 \le j \le 3$) department is $a_{i,j}$. The satisfaction of an assignment is defined as the sum, over all new members, of their satisfaction with the department they are assigned to. That is, if the i-th ($1 \le i \le n$) new member is assigned to the d_i-th department, where $d_i \in \{1, 2, 3\}$, then the satisfaction of the assignment is $\sum_{i=1}^{n} a_{i,d_i}$.

Little L does not want any single department to have too many new members. Specifically, he requires that in the assignment there is no department that is assigned more than $\frac{n}{2}$ new members. You need to help Little L find the maximum satisfaction of an assignment that satisfies his requirement.

Input

Read the data from the file `club.in`.

This problem contains multiple test cases.

The first line of the input contains a positive integer t, the number of test cases.

Then the test cases follow one after another. For each test case:

- The first line contains a positive integer n, the number of new members.
- Line $i + 1$ ($1 \le i \le n$) contains three non-negative integers $a_{i,1}, a_{i,2}, a_{i,3}$, the satisfaction of the i-th new member with departments $1, 2, 3$ respectively.

Output

Write the output to the file `club.out`.

For each test case, output one line containing one non-negative integer, the maximum satisfaction of an assignment that satisfies Little L's requirement.

Sample 1 input

```
3
4
4 2 1
3 2 4
5 3 4
3 5 1
4
0 1 0
0 1 0
0 2 0
0 2 0
```

```
2
10 9 8
4 0 0
```

Sample 1 output

```
18
4
13
```

Explanation of sample 1

This sample contains three test cases.

For the first test case, the four new members can be assigned to departments $1, 3, 1, 2$ respectively. Then the numbers of new members in the three departments are $2, 1, 1$, none exceeding $\frac{4}{2} = 2$, and the satisfaction is $4 + 4 + 5 + 5 = 18$.

For the second test case, the four new members can be assigned to departments $1, 1, 2, 2$ respectively. Then the numbers of new members in the three departments are $2, 2, 0$, none exceeding $\frac{4}{2} = 2$, and the satisfaction is $0 + 0 + 2 + 2 = 4$.

For the third test case, the two new members can be assigned to departments $2, 1$ respectively. Then the numbers of new members in the three departments are $1, 1, 0$, none exceeding $\frac{2}{2} = 1$, and the satisfaction is $9 + 4 = 13$.

Sample 2

See `club/club2.in` and `club/club2.ans` in the contestant's directory.

This sample satisfies the constraints of test cases $3, 4$.

Sample 3

See `club/club3.in` and `club/club3.ans` in the contestant's directory.

This sample satisfies the constraints of test cases $5 \sim 8$.

Sample 4

See `club/club4.in` and `club/club4.ans` in the contestant's directory.

This sample satisfies the constraints of test case 9.

Sample 5

See `club/club5.in` and `club/club5.ans` in the contestant's directory.

This sample satisfies the constraints of test cases $15, 16$.

Constraints

For all test data, it is guaranteed that:

- $1 \le t \le 5$;
- $2 \le n \le 10^5$, and n is even;

- for all $1 \le i \le n$, $1 \le j \le 3$, we have $0 \le a_{i,j} \le 2 \times 10^4$.

Test case no.	$n =$	Special property
1	2	None
2	4	
$3, 4$	10	
$5 \sim 8$	30	
9	200	B
$10, 11$		None
12	10^5	A
$13, 14$		B
$15, 16$		C
$17 \sim 20$		None

Special property A: for all $1 \le i \le n$, we have $a_{i,2} = a_{i,3} = 0$.

Special property B: for all $1 \le i \le n$, we have $a_{i,3} = 0$.

Special property C: for all $1 \le i \le n$, $1 \le j \le 3$, each $a_{i,j}$ is generated independently and uniformly at random in $[0, 2 \times 10^4]$.

Problem 2: Road Repair (道路修复, `road`) QOJ 14598

Time limit: 1 s. Memory limit: 512 MB. Total points: 100.

Problem description

The transportation system of Country C consists of n cities and m bidirectional roads, each connecting two cities. The i-th ($1 \le i \le m$) road connects cities u_i and v_i. Any two cities can reach each other through some of the roads.

However, recently, because of a major earthquake, all m roads were destroyed, and the cost of repairing the i-th ($1 \le i \le m$) road is w_i. At the same time, Country C also has k towns that are prepared to undergo urbanization. For the j-th ($1 \le j \le k$) town, the cost for Country C to urbanize it is c_j. After the j-th ($1 \le j \le k$) town has been urbanized, several roads may be built between this town and the original n cities, where the cost of building a road between it and the i-th ($1 \le i \le n$) city is $a_{j,i}$. Country C may choose any number of these k towns to urbanize, and may also choose not to urbanize any town.

To restore the traffic between the cities as soon as possible, the government of Country C wants to connect the original n cities pairwise at the lowest cost, that is, any two of the original cities can reach each other through some of the repaired or newly built roads. You need to help them find the minimum cost of connecting the original n cities pairwise.

Input

Read the data from the file `road.in`.

The first line of the input contains three non-negative integers n, m, k, the number of original cities, the number of roads, and the number of towns prepared to undergo urbanization, respectively.

Line $i+1$ ($1 \le i \le m$) of the input contains three non-negative integers u_i, v_i, w_i, the two cities connected by the i-th road and the cost of repairing this road.

Line $j+m+1$ ($1 \le j \le k$) of the input contains $n+1$ non-negative integers $c_j, a_{j,1}, a_{j,2}, \ldots, a_{j,n}$, the cost of urbanizing the j-th town and the costs of building roads between this town and the original cities, respectively.

Output

Write the output to the file `road.out`.

Output one line containing one non-negative integer, the minimum cost of connecting the original n cities pairwise.

Sample 1 input

```
4 4 2
1 4 6
2 3 7
4 2 5
4 3 4
1 1 8 2 4
100 1 3 2 4
```

Sample 1 output

```
13
```

Explanation of sample 1

The government of Country C can choose to repair the 3rd and 4th roads, then urbanize the 1st town and build the roads between it and cities $1, 3$. The total cost is $5+4+1+1+2=13$. It can be shown that there is no cost smaller than 13 that connects the original 4 cities pairwise.

Sample 2

See `road/road2.in` and `road/road2.ans` in the contestant's directory.

This sample satisfies the constraints of test cases $11, 12$.

Sample 3

See `road/road3.in` and `road/road3.ans` in the contestant's directory.

This sample satisfies the constraints of test cases $13, 14$.

Sample 4

See `road/road4.in` and `road/road4.ans` in the contestant's directory.

This sample satisfies the constraints of test cases $15, 16$.

Constraints

For all test data, it is guaranteed that:

- $1 \le n \le 10^4$, $1 \le m \le 10^6$, $0 \le k \le 10$;
- for all $1 \le i \le m$, we have $1 \le u_i, v_i \le n$, $u_i \ne v_i$ and $0 \le w_i \le 10^9$;
- for all $1 \le j \le k$, we have $0 \le c_j \le 10^9$;
- for all $1 \le j \le k$, $1 \le i \le n$, we have $0 \le a_{j,i} \le 10^9$;
- any two of the original cities can reach each other through some of the original roads.

<table>
<tr><th>Test case no.</th><th>$n \le$</th><th>$m \le$</th><th>$k \le$</th><th>Special property</th></tr>
<tr><td>$1 \sim 4$</td><td>10^4</td><td>10^6</td><td>0</td><td>None</td></tr>
<tr><td>$5, 6$</td><td rowspan="6">10^3</td><td>10^5</td><td rowspan="4">5</td><td>A</td></tr>
<tr><td>$7, 8$</td><td>None</td></tr>
<tr><td>$9, 10$</td><td rowspan="4">10^6</td><td>A</td></tr>
<tr><td>$11, 12$</td><td>None</td></tr>
<tr><td>$13, 14$</td><td rowspan="2">10</td><td>A</td></tr>
<tr><td>$15, 16$</td><td>None</td></tr>
<tr><td>$17, 18$</td><td rowspan="3">10^4</td><td rowspan="2">5</td><td>A</td></tr>
<tr><td>$19, 20$</td><td rowspan="2">None</td></tr>
<tr><td>$21 \sim 25$</td><td>10</td></tr>
</table>

Special property A: for all $1 \le j \le k$, we have $c_j = 0$, and there exists $1 \le i \le n$ such that $a_{j,i} = 0$.

Problem 3: Homophonic Replacement (谐音替换, `replace`) QOJ 14599

Time limit: 1 s. Memory limit: 2048 MB. Total points: 100.

Problem description

Little W is a competitive programmer who likes linguistics. In linguistics, homophonic replacement means replacing original words by words with the same or a similar pronunciation. Little W found that the process of homophonic replacement can be described with strings. Specifically, Little W defines homophonic replacement as the following string problem.

Given n pairs of strings, the i-th ($1 \le i \le n$) pair of strings is $(s_{i,1}, s_{i,2})$, satisfying $|s_{i,1}| = |s_{i,2}|$, where $|s|$ denotes the length of the string s.

For a string s, a replacement of s is defined as follows:

- For some substring y of s, if there exists $1 \le i \le n$ such that $y = s_{i,1}$, then replace y by $y' = s_{i,2}$. Specifically, let $s = x + y + z$, where x and z may be empty and "+" denotes string concatenation; then the replacement of s yields the string $s' = x + y' + z$.

Little W poses q questions. The j-th ($1 \le j \le m$) question gives two distinct strings $t_{j,1}, t_{j,2}$, and she wants to know how many replacements of the string $t_{j,1}$ yield the string $t_{j,2}$. Two replacements of s are different if and only if the position of the substring y is different or the pair $(s_{i,1}, s_{i,2})$ used for the replacement is different, that is, x, z are different or i is different. You need to answer all the questions posed by Little W.

Input

Read the data from the file `replace.in`.

The first line of the input contains two positive integers n, q, the number of pairs of strings and the number of questions posed by Little W, respectively.

Line $i + 1$ ($1 \le i \le n$) of the input contains two strings $s_{i,1}, s_{i,2}$, the i-th pair of strings.

Line $j + n + 1$ ($1 \le j \le q$) of the input contains two strings $t_{j,1}, t_{j,2}$, the j-th question posed by Little W.

Output

Write the output to the file `replace.out`.

Output q lines, where the j-th ($1 \le j \le q$) line contains one non-negative integer, the number of replacements of the string $t_{j,1}$ that yield the string $t_{j,2}$.

Sample 1 input

```
4 2
xabcx xadex
ab cd
bc de
aa bb
xabcx xadex
aaaa bbbb
```

Sample 1 output

```
2
0
```

Explanation of sample 1

For Little W's first query, there are 2 replacements of $t_{1,1}$ that yield $t_{1,2}$:

1. Let x, z both be the empty string, $y =$ `xabcx`, $i = 1$; then $y' =$ `xadex`, and the replacement yields `xadex`;
2. Let $x =$ `xa`, $y =$ `bc`, $z =$ `x`, $i = 3$; then $y' =$ `de`, and the replacement yields `xadex`.

Sample 2 input

```
3 4
a b
b c
c d
aa bb
aa b
a c
b a
```

Sample 2 output

```
0
0
0
0
```

Sample 3

See `replace/replace3.in` and `replace/replace3.ans` in the contestant's directory.
This sample satisfies the constraints of test cases 11, 12.

Sample 4

See `replace/replace4.in` and `replace/replace4.ans` in the contestant's directory.
This sample satisfies the constraints of test cases 15, 16.

Constraints

Let $L_1 = \sum_{i=1}^{n} |s_{i,1}| + |s_{i,2}|$ and $L_2 = \sum_{j=1}^{q} |t_{j,1}| + |t_{j,2}|$. For all test data, it is guaranteed that:

- $1 \le n, q \le 2 \times 10^5$;
- $2 \le L_1, L_2 \le 5 \times 10^6$;
- for all $1 \le i \le n$, $s_{i,1}, s_{i,2}$ consist only of lowercase English letters, and $|s_{i,1}| = |s_{i,2}|$;

- for all $1 \le j \le q$, $t_{j,1}, t_{j,2}$ consist only of lowercase English letters, and $t_{j,1} \neq t_{j,2}$.

Test case no.	$n, q \le$	$L_1, L_2 \le$	Special property
$1, 2$	10^2	200	None
$3 \sim 5$	10^3	2,000	
6		10^6	AB
$7, 8$	10^4		A
$9, 10$	2×10^5		B
$11, 12$		2×10^6	None
$13, 14$		5×10^6	A
$15, 16$			B
$17 \sim 20$			None

Special property A: $q = 1$.

Special property B: define a string s to be special if and only if s consists only of the characters `a` and `b`, and the character `b` occurs exactly once in s. For all $1 \le i \le n$, $s_{i,1}, s_{i,2}$ are special, and for all $1 \le j \le q$, $t_{j,1}, t_{j,2}$ are special.

Problem 4: Employee Recruitment (员工招聘, `employ`) QOJ 14600

Time limit: 1 s. Memory limit: 512 MB. Total points: 100.

Problem description

Little Z and Little H want to start a company together. A total of n people, numbered $1 \sim n$, have come to apply. Little Z and Little H want to hire at least m people.

Little H is the interviewer and will interview one person per day over the next n days. Little Z is in charge of deciding the order in which the applicants come for their interviews. Specifically, Little Z may choose a permutation p of $1 \sim n$, and then on day i ($1 \le i \le n$) notify the person numbered p_i to come for an interview.

Little H has prepared n sets of interview questions of varying difficulty. Since the n applicants are of roughly the same ability, for the same set of questions, the results of all people are identical. Specifically, the difficulty of the interview questions on day i ($1 \le i \le n$) is $s_i \in \{0, 1\}$, where $s_i = 0$ means that this set is hard and nobody can solve it, and $s_i = 1$ means that this set is easy and everybody can solve it. Little H decides whether to hire according to the interviewee's result, that is, if the interviewee does not solve the questions, they are rejected; otherwise they are hired.

However, everybody's patience has a certain limit; if too many people were not hired before their own interview, they will simply give up on attending the interview. Specifically, the patience limit of the person numbered i ($1 \le i \le n$) is described by a non-negative integer c_i: if at least c_i people before them have been rejected or have given up on attending the interview, then they will also give up on attending the interview.

Little Z wants to know how many interview orders p in total allow them to hire at least m people. You need to help Little Z find the number of permutations p for which at least m people are hired. Since the answer may be large, you only need to find the answer modulo 998,244,353.

Input

Read the data from the file `employ.in`.

The first line of the input contains two positive integers n, m, the number of applicants and the number of people they want to hire, respectively.

The second line of the input contains a string $s_1 \dots s_n$ of length n, the difficulty of the interview questions on each day.

The third line of the input contains n non-negative integers $c_1, c_2, \dots, c_n$, the patience limit of each person.

Output

Write the output to the file `employ.out`.

Output one line containing one non-negative integer, the number of permutations p for which at least m people are hired, modulo 998,244,353.

Sample 1 input

```
3 2
101
1 1 2
```

Sample 1 output

```
2
```

Explanation of sample 1

There are the following 2 interview orders p that allow Little Z and Little H to hire at least 2 people:

1. $p = [1, 2, 3]$, hiring the person numbered 1 and then the person numbered 3;
2. $p = [2, 1, 3]$, hiring the person numbered 2 and then the person numbered 3.

Sample 2 input

```
10 5
1101111011
6 0 4 2 1 2 5 4 3 3
```

Sample 2 output

```
2204128
```

Sample 3

See `employ/employ3.in` and `employ/employ3.ans` in the contestant's directory.
This sample satisfies the constraints of test cases $6 \sim 8$.

Sample 4

See `employ/employ4.in` and `employ/employ4.ans` in the contestant's directory.
This sample satisfies the constraints of test cases $12 \sim 14$.

Sample 5

See `employ/employ5.in` and `employ/employ5.ans` in the contestant's directory.
This sample satisfies the constraints of test cases $18 \sim 21$.

Constraints

For all test data, it is guaranteed that:

- $1 \leq m \leq n \leq 500$;
- for all $1 \leq i \leq n$, we have $s_i \in \{0, 1\}$;
- for all $1 \leq i \leq n$, we have $0 \leq c_i \leq n$.

Test case no.	$n \leq$	m	Special property
$1, 2$	10	$\leq n$	None
$3 \sim 5$	18		
$6 \sim 8$	10^2		A
$9 \sim 11$			
$12 \sim 14$	500	$= 1$	None
15		$= n$	
$16, 17$		$\leq n$	A
$18 \sim 21$			B
$22 \sim 25$			None

Special property A: for all $1 \leq i \leq n$, we have $s_i = 1$.
Special property B: among $s_1, s_2, \ldots, s_n$ at most 18 take the value 1, that is, $\sum_{i=1}^{n} s_i \leq 18$.

Source

CSP-S 2025 (CCF CSP-J/S 2025 second round, Senior level), held 2025-10-25. The statements were taken from the official statement PDFs embedded on the QOJ problem pages. Contest page: `https://qoj.ac/contest/2571`. Problem pages:

- Problem 1: `https://qoj.ac/problem/14597`
- Problem 2: `https://qoj.ac/problem/14598`
- Problem 3: `https://qoj.ac/problem/14599`
- Problem 4: `https://qoj.ac/problem/14600`